

Replica-synthesized molecular dynamics in the browser

Direct validation of a STORMM-inspired WebGPU engine in the Marco molecular modeler

ABSTRACT

We implemented an isolated-boundary molecular-mechanics engine in WebGPU for the Marco browser molecular modeler, adopting STORMM's strategy of advancing many independent systems in one GPU synthesis and fixed-point, order-independent accumulation. Direct comparisons used upstream STORMM commit 3eecbde5192. Alanine dipeptide (22 atoms) and Trp-cage (304 atoms) passed preregistered energy and all-force thresholds for every supported vacuum term and OBC2 generalized Born. Sixteen-replica NVE tests, bitwise replay, replica isolation, and replica-local SHAKE/RATTLE constraints also passed. On one NVIDIA L4, Marco reached 0.51x and 0.73x upstream CUDA throughput in flexible vacuum and OBC2 simulations, and 0.39x and 0.52x with 12 X-H constraints at 2 fs. The study establishes numerical agreement and useful performance for a deliberately limited browser engine; it does not establish feature parity with STORMM.

Key result

A reproducible browser engine reaches 39-73% of upstream STORMM CUDA throughput for the tested 16-replica L4 workloads while preserving force agreement, deterministic isolation, and constrained dynamics.

1 Motivation

STORMM combines many molecular systems into a synthesis to fill a GPU efficiently and uses fixed-precision representations to make parallel accumulation reproducible. This architecture is attractive in a browser, where one GPU command stream can replace repeated JavaScript-to-device round trips and users benefit from local, installation-free simulation. The original STORMM paper showed that batching many small implicit-solvent simulations can outperform independent processes using NVIDIA MPS [1].

The present question is narrower: can a browser WebGPU implementation reproduce upstream STORMM's energy surface and forces for the terms it claims to implement, while preserving independent replica behavior and useful small-system throughput?

2 Implementation

The engine evaluates harmonic bonds and angles, cosine periodic torsions, isolated no-cutoff Coulomb and 12-6 Lennard-Jones interactions with explicit exceptions, and OBC2 generalized Born with an optional ACE surface term. It accepts numeric pair-distance constraints; Marco derives X-H targets from equilibrium bonds and selects a 2 fs time step.

One WebGPU workgroup advances each replica. Deterministic Gauss-Seidel SHAKE follows drift, and RATTLE projects velocities after the second kick. Contributions use split signed 64-bit fixed-point arithmetic: 2^{22} units per kcal/mol for energy, 2^{18} units per kcal/mol/Å for force, and 2^{32} units per Å for authoritative coordinates. Rounded coordinate and constraint increments return to that fixed-point representation. The same seed produces bitwise-identical trajectories.

Marco normally converts a complete numeric OpenMM System after browser parameterization, without repeating chemical perception. For this study, AMBER topologies supplied by STORMM were exported to the same numeric schema. Validation metadata selects STORMM's AMBER Coulomb constant and matched OBC2 settings. This implementation is STORMM-inspired, not a source translation; it uses no upstream CUDA kernels or data structures.

3 Validation design

3.1 Static energies and forces

Alanine dipeptide and Trp-cage coordinates and AMBER topologies came from upstream STORMM. Native references were extracted from committed potential regression tests and snapshot arrays. Bonds, angles, periodic torsions, Coulomb (ordinary plus 1-4), and Lennard-Jones (ordinary plus 1-4) were tested separately and as a supported-term total. OBC2 used external dielectric 78.5 with the surface-area term disabled to match the upstream polar-only snapshot.

Quantity	Pass rule
Energy	absolute error ≤ 0.02 kcal/mol or relative error $\leq 5 \times 10^{-4}$
Force	relative RMS $\leq 2 \times 10^{-3}$ or absolute RMS $\leq 5 \times 10^{-4}$ kcal/mol/Å
NVE drift	maximum normalized windowed drift $\leq 0.5\%$
Constraint distance	maximum relative residual $\leq 2 \times 10^{-5}$

Table 1. Preregistered acceptance criteria encoded in the test programs.

Alanine's one CMAP term (-0.38861139 kcal/mol) is not implemented; it is recorded but excluded from the supported total. Trp-cage requires no omitted term.

3.2 Dynamics and constraints

Sixteen alanine replicas were propagated for 5,000 unconstrained NVE steps at 1 fs in vacuum and matched OBC2. Windowed energy drift was normalized by $3Nk_B T/2$ at 300 K. Two identical four-replica Langevin runs tested bitwise replay; a one-replica run and replica zero of a 16-replica synthesis tested cross-replica isolation.

Constraint tests used four independent stacks of eight water molecules, with 16 coupled O-H constraints per replica, for 500 Langevin steps at 2 fs and 1×10^{-5} relative tolerance. Tests covered distance and velocity residuals, replay, one-versus-four replica isolation, and an intentional iteration-limit failure. A separate 500-step water test compared Marco SHAKE/RATTLE with OpenMM Reference/CCMA.

3.3 Performance

The matched workload contained 16 alanine replicas and 10,000 steps in vacuum or polar-only OBC2. Flexible runs used 1 fs. Constrained runs used 12 X-H distances and 2 fs; upstream STORMM used `rigid_geom` on. Each median contains five repeats. Upstream `dynamics.stormm.cuda` and WebGPU/Vulkan ran sequentially on one licensed NVIDIA L4 (driver 595.71.05, CUDA 12.9). Native STORMM used its default single-precision valence and nonbonded settings.

4 Results

4.1 Direct agreement with native STORMM snapshots

System	Atoms	Max energy rel. error	Total force rel. RMS	OBC2 energy rel. error	OBC2 force rel. RMS
Alanine dipeptide	22	5.09×10^{-6}	4.30×10^{-4}	2.24×10^{-7}	7.48×10^{-7}
Trp-cage	304	3.06×10^{-6}	2.34×10^{-5}	1.60×10^{-6}	3.07×10^{-6}

Table 2. Same-L4 WebGPU errors against pinned native STORMM snapshots. All checks passed.

The maximum absolute supported-total force error was 6.20×10^{-4} kcal/mol/Å for alanine and 1.10×10^{-3} kcal/mol/Å for Trp-cage. The separately maintained OpenMM Reference suite also passed for all existing fixtures, including OBC2/ACE.

4.2 Dynamics and constraints

Model	Maximum normalized windowed drift	Median across 16 replicas
Vacuum	0.2786%	0.0630%
OBC2 polar-only	0.2748%	0.0607%

Table 3. Five-thousand-step NVE conservation at 1 fs on the L4.

Deterministic replay produced bitwise-identical energies and positions. Replica zero was also bitwise identical when advanced alone or as part of a 16-system synthesis.

SHAKE distance	2.80×10^{-7}	RATTLE velocity	2.78×10^{-8}
Marco vs OpenMM	9.67×10^{-6}	OpenMM CCMA	3.50×10^{-7}

Constraint residuals. Marco and OpenMM maximum relative distance errors both passed the shared 2×10^{-5} criterion; replay and replica isolation were bitwise exact.

4.3 Same-GPU throughput

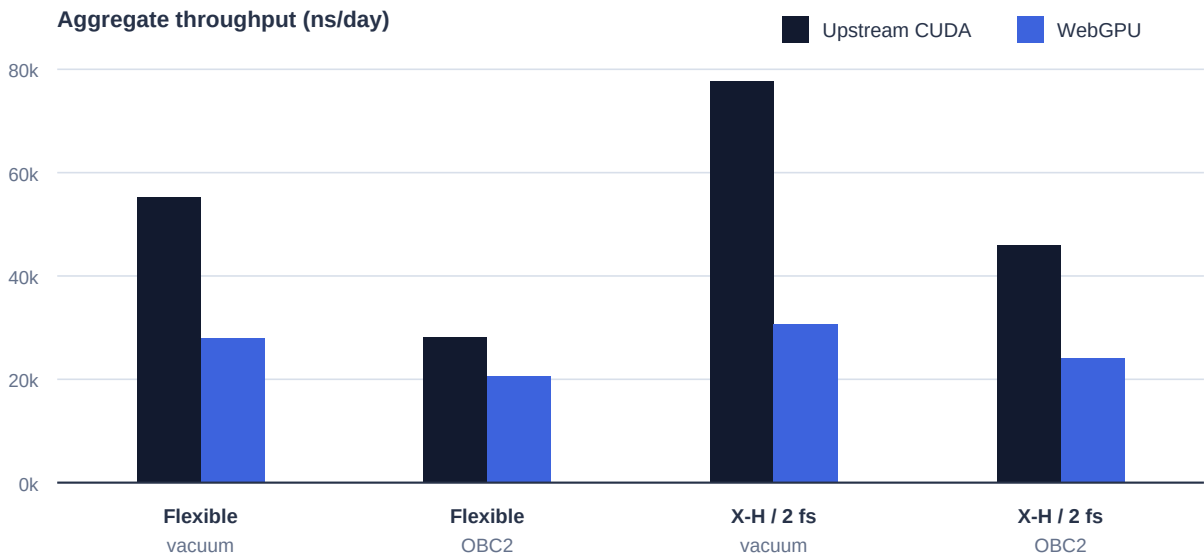


Figure 1. Median aggregate throughput for 16 alanine replicas and 10,000 steps. Both executables ran sequentially on the same NVIDIA L4; five repeats per condition.

Dynamics	Model	STORMM CUDA ns/day	Marco WebGPU ns/day	WebGPU / CUDA
Flexible, 1 fs	Vacuum	55,340.3	28,046.9	0.507x
Flexible, 1 fs	OBC2 polar-only	28,069.0	20,564.8	0.733x
X-H constrained, 2 fs	Vacuum	77,728.4	30,635.2	0.394x
X-H constrained, 2 fs	OBC2 polar-only	46,026.3	24,016.4	0.522x

Table 4. Exact values plotted in Figure 1.

For this replicated workload, Marco reaches 51-73% of upstream CUDA throughput without constraints and 39-52% with constraints. The larger constrained gap is consistent with Marco's general serial-per-replica Gauss-Seidel solver versus specialized native constraint machinery; constraint parallelization is therefore a concrete optimization target.

4.4 Auxiliary laptop result

On an M1 Pro, Marco WebGPU reached 12,009.1 ns/day in vacuum and 8,518.1 ns/day in OBC2, versus 10,460.8 and 4,231.5 ns/day for a native CPU executable configured explicitly for double precision. These values demonstrate local browser execution but are not used for the primary speed claim because devices and precision modes differ.

5 Limitations and defensible claim

The browser engine is nonperiodic and all-pairs. It supports general numeric pair-distance constraints and X-H SHAKE/RATTLE, but not specialized SETTLE or parallel constraint-graph kernels. It lacks PME, periodic boundaries, neighbor lists, CMAP, GBn/GBn2, virtual sites, heterogeneous topologies within one synthesis, and free-energy methods. Marco currently caps a replica at 512 atoms. Production OBC2 uses external dielectric 78.3 and ACE; the polar-only 78.5 configuration exists solely for exact upstream matching.

For the listed isolated-boundary terms and two pinned AMBER fixtures, the Marco WebGPU engine reproduces upstream STORMM regression energies and forces within stated tolerances, preserves deterministic independent replicas with or without X-H constraints, and reaches 39-73% of upstream STORMM CUDA throughput for the tested 16-replica NVIDIA L4 workloads.

It is not yet defensible to call this a full STORMM port or to extrapolate throughput to large periodic biomolecular simulations. The original paper's ff14SB side-chain-only, GBn, hydrogen-mass-repartitioned 4 fs workload is intentionally not treated as a direct comparator.

6 Reproducibility

Inputs, pinned source commit, source and executable hashes, complete force vectors, encoded thresholds, repeat timings, hardware labels, and result hashes are stored under `stormm-study`. Accepted same-L4 manifests are in `results/l4-vws`. The upstream source required one recorded compile-only member-access patch that changes no molecular-mechanics equation. The L4 runner pins the NVIDIA Vulkan ICD, rejects software adapters, runs native CUDA and WebGPU sequentially, records the environment, and hashes all outputs.

7 References

- [1] D. S. Cerutti, R. Wiewiora, S. Boothroyd, and W. Sherman. "STORMM: Structure and topology replica molecular mechanics for chemical simulations." *Journal of Chemical Physics* 161, 032501 (2024). doi:10.1063/5.0211032.
- [2] STORMM developers. Official source repository, pinned at commit 3eecbde519254620fdc024f396fa07df78ea69e. github.com/psivant/stormm.
- [3] STORMM developers. Molecular Dynamics in STORMM. stormm.psithera.com/stormm-md.

Software availability: the validation package, browser engine, accepted manifests, and regeneration commands accompany this working preprint in the Marco repository.